Code for multimode fiber scrambled image reconstruction using optical diffractive neural networks

Haoyi Yu[1,2], Zihao Huang[1,2], Min Gu[1,2,3*]

1. School of Artificial Intelligence Science and Technology, University of Shanghai for Science and Technology, Shanghai 200093, China
2. Institute of Photonic Chips, University of Shanghai for Science and Technology, Shanghai 200093, China
3. Zhangjiang Laboratory, Shanghai 201210, China

Email: gumin@usst.edu.cn

This document is designated to describe the code for multimode fiber scrambled image reconstruction using optical diffractive neural networks (DN_2s).

Code section:

In the code section, there are: one main training code (**train_01.py**), and one calculation model configuration setup code (**config.json**), and several basic utilized codes.

The physics model of DN_2s is located in the file folder named "models", and with a name of "**Onn_Net.py**". In this model, the wave propagation model is described, with each layer of the DN_2s are defined based on the Rayleigh Sommerfeld equation and fast Fourier transform (FFT) equations. Layer distance, diffractive neuron size can be directly modified from this file.

To set up the modeling, the parameters of the calculation, one need to reset them in the **config.json** file, including wavelength (lamda), pixel number, layer number. The learning rate is also listed.

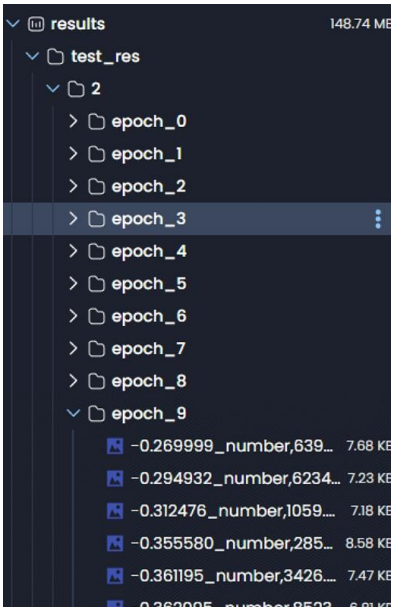
Data section

10000 pairs of speckled images (amplitude and phase) and their ground truth images are presented in this demonstration. 8000 of them are used in the training process, and 2000 of them are used in the testing process. The training data and the testing data are randomly separated to ensure that reliable training and testing results can be achieved. These data is located in the "**data**" section, under the file folder of "**image**".

Training and testing process demonstration

Real time loss values during training and testing process can be observed in the command window, and it looks like the screen shot below. The loss (NPCC) values for each image used in the training process can be seen during the training process.

```
Run 816598/out...
2024-11-05 14:25:50,403 - train_01.py [line:91] - INFO: Test Data: 872 Loss: -0.598549
2024-11-05 14:25:50,407 - train_01.py [line:91] - INFO: Test Data: 873 Loss: -0.650949
2024-11-05 14:25:50,411 - train_01.py [line:91] - INFO: Test Data: 874 Loss: -0.603289
2024-11-05 14:25:50,415 - train_01.py [line:91] - INFO: Test Data: 875 Loss: -0.736875
2024-11-05 14:25:50,419 - train_01.py [line:91] - INFO: Test Data: 876 Loss: -0.518512
2024-11-05 14:25:50,422 - train_01.py [line:91] - INFO: Test Data: 877 Loss: -0.764205
2024-11-05 14:25:50,426 - train_01.py [line:91] - INFO: Test Data: 878 Loss: -0.613831
2024-11-05 14:25:50,430 - train_01.py [line:91] - INFO: Test Data: 879 Loss: -0.647123
2024-11-05 14:25:50,434 - train_01.py [line:91] - INFO: Test Data: 880 Loss: -0.697985
2024-11-05 14:25:50,438 - train_01.py [line:91] - INFO: Test Data: 881 Loss: -0.645791
2024-11-05 14:25:50,441 - train_01.py [line:91] - INFO: Test Data: 882 Loss: -0.570758
2024-11-05 14:25:50,445 - train_01.py [line:91] - INFO: Test Data: 883 Loss: -0.601819
```

Results section

To save memory space, in this demonstration, only 10 epoches are used for a 3 layers DN_2s at the wavelength of 532 nm. The results are saved in the results section, with file folder named “test_res”. Under this folder, one can find the test results for each epoch. Each test result is named with the format showing the value of NPCC and label number. This is shown in the screen shot blew.